GX140 3in ANALOG FPV — ELECTRICAL SCHEMATIC (Rev 2.0)

METHOD: point-to-point by net label. Every conductor breaks out at the device pin.
Match net names to find the far end. No routed runs, no crossings, no junction ambiguity.

=== POWER TREE ===

BAT XT30 -> ESC VBAT pads (22AWG). FC powered via ESC 8-pin harness (VBAT/GND/CURR/S1-S4).
FC internal 5V/3A BEC -> 5V net -> VTX, CAM, RX, GPS.

=== UART MAP (F405 Mini) ===

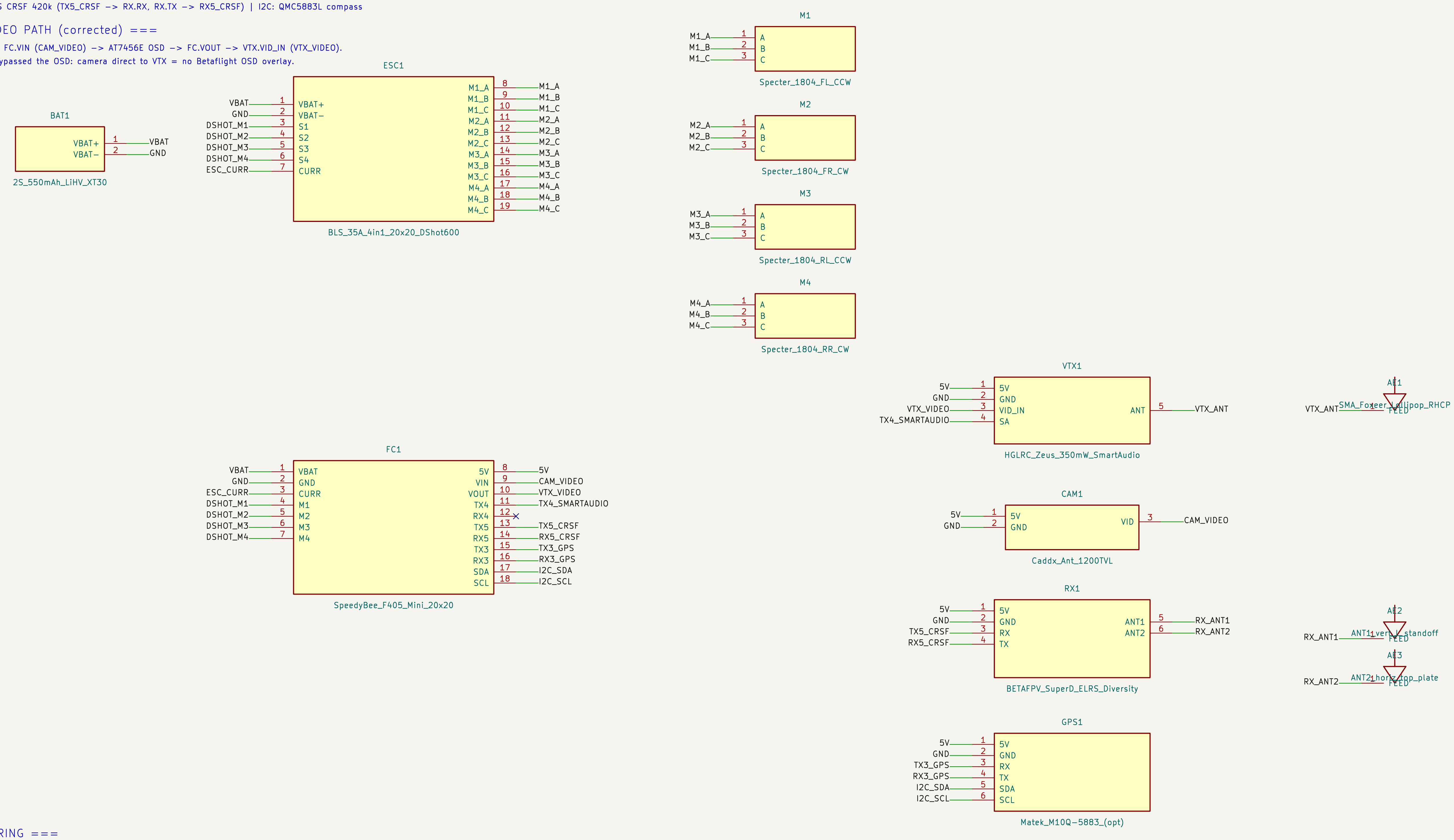
UART3: GPS 57600 (TX3_GPS / RX3_GPS) | UART4: SmartAudio 2.1 (TX4 only, RX4 = N/C)
UART5: ELRS CRSF 420k (TX5_CRSF -> RX.RX, RX.TX -> RX5_CRSF) | I2C: QMC5883L compass

=== VIDEO PATH (corrected) ===

CAM.VID -> FC.VIN (CAM_VIDEO) -> AT7456E OSD -> FC.VOUT -> VTX.VID_IN (VTX_VIDEO).
Grok rev1 bypassed the OSD: camera direct to VTX = no Betaflight OSD overlay.

MOTOR PHASE WIRING: 3 conductors per motor (A/B/C).

Swap any two phases to reverse direction. M1 FL CCW | M2 FR CW | M3 RL CCW | M4 RR CW.



=== WIRING ===

22AWG: VBAT, GND, motor phase | 26AWG: UART, DShot, video | 28AWG: I2C
JST-SH 1.0mm: CAM | JST-GH 1.25mm: GPS | XT30: battery | Heat shrink all joints

=== SAFETY ===

Smoke stopper (ShortSaver) MANDATORY first power-up | Props OFF for motor direction test
Continuity check VBAT-GND before battery | Verify motor order/direction in Betaflight Motors tab

VTX: Zeus 350mW | CAM: Caddx Ant | RX: SuperD ELRS | GPS: M10Q-5883 (opt)
Motors: HGLRC Specter 1804 3500KV | Props: GEMFAN 3052
FC: SpeedyBee F405 Mini | ESC: BLS 35A 4in1 DShot600
Frame: GX140 140mm twin 20x20, 4mm carbon | AUW ~190g
GX140 Build — net-label point-to-point method

Sheet: /
File: gx140_rev2.kicad_sch

Title: GX140 3in Analog FPV — Electrical Schematic

Size: A3	Date: 2026-06-12	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1